Self-Supervised Mamba for Hyperspectral Image Classification

Minghao Zhu, Heng Wang, Yuebo Meng, Shengjun Xu, Yaohai Lin, Zhe Shan, and Zongfang Ma

***Abstract*—Recent advances in Mamba-based architectures have demonstrated promising potential for hyperspectral image classification (HSIC), offering linear-complexity long-range dependency modeling. However, two critical challenges persist in adapting this paradigm to hyperspectral image (HSI) analysis: the substantial requirement for annotated training samples and insufficient capacity to interpret the intricate spatial–spectral features inherent in HSI data, particularly under few-shot learning (FSL) scenarios. To address these limitations, we present SSupMamba, a novel self-supervised Mamba framework tailored for HSIC. First, we propose a composite scanning Mamba block (CSMB) that enables comprehensive global feature extraction through multidirectional selective scanning of HSI data cubes. Second, we develop a spatial-spectral masked Mamba (SAEM) framework that uses randomized masking and reconstruction tasks to enhance local representation learning. Third, we establish a unified self-supervised architecture incorporating contrastive learning (CL) to maximize mutual information between multiviews while preserving intrinsic spatial–spectral characteristics. Experimental results on four public datasets demonstrate that the proposed method exhibits excellent feature extraction capabilities under few-shot conditions and outperforms several state-of-the-art HSIC methods. The code is available at: https://github.com/Winkness/SSupMamba.**

***Index Terms*—Contrastive learning (CL), few-shot learning (FSL), hyperspectral image classification (HSIC), masked image modeling (MIM), self-supervised learning (SSL), spatial–spectral Mamba.**

I. Introduction

HYPERSPECTRAL image (HSI) has revolutionized remote sensing by capturing contiguous spectral signatures across hundreds of narrow wavelength bands, enabling unprecedented material discrimination capabilities for land cover analysis [1], [2]. As a pivotal task in HSI interpretation, hyperspectral image classification (HSIC) aims to assign precise semantic labels to each pixel through joint analysis of spatial–spectral features. This technique drives critical applications in environmental monitoring [3], agricultural optimization [4], and defense systems [5]. Nevertheless, four inherent challenges persist: the dimensionality curse stemming from high spectral redundancy, data scarcity of annotated samples, class imbalance in real-world scenarios, and complex spatial–spectral interactions in heterogeneous landscapes [6].

Substantial research efforts have been devoted to developing HSIC methods. Early spectral-based approaches [7], [8] prioritized band analysis but lacked structural feature extraction. The advent of deep learning has shifted the paradigm to automated feature learning via convolutional neural networks (CNNs) [9], [10], graph convolutional networks (GCNs) [11], [12], and vision transformers (ViTs) [13], [14]. Subsequent variants [15], [16], [17], [18] further enhanced model performance. While CNNs demonstrate local feature extraction capabilities, their limited receptive fields hinder long-range dependency modeling. Although ViTs overcome this limitation through global self-attention mechanisms, their quadratic computational complexity ($O(N^2)$) becomes prohibitive for high-resolution HSI processing [19], [20].

The recent emergence of state-space models (SSMs), particularly the Mamba architecture [19], presents a paradigm shift through its selective state-space mechanism (S6). Mamba achieves linear computational scaling ($O(N)$) while maintaining global context awareness, demonstrating superior efficiency in natural language processing (NLP) [21], [22] and computer vision (CV) [23], [24]. Initial applications in HSIC [25], [26] validate Mamba's potential but reveal critical limitations: 1) dependence on large training sets conflicting with real-world HSI annotation constraints; 2) inadequate handling of complex spatial–spectral interactions in irregular geospatial patterns (e.g., meandering rivers, fragmented urban areas) through existing directional scanning strategies; and 3) underutilization of self-supervised learning to mitigate data scarcity. These shortcomings underscore the need for innovative approaches combining sample-efficient learning with advanced spectral–spatial modeling.

We address these challenges through a self-supervised Mamba framework that synergizes masked image modeling (MIM) and contrastive learning (CL). Our methodological innovations stem from three key observations: First, MIM's

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reconstruction-driven paradigm [27], [28] naturally aligns with HSI's structural redundancy, enabling efficient feature learning from limited data. Second, CL's instance discrimination capability [29], [30] complements MIM by enhancing interclass separability. Third, Mamba's directional scanning mechanism can be reengineered to capture noncausal spatial–spectral relationships through multidirectional processing.

Building on these insights, we propose SSupMamba—the first self-supervised Mamba architecture specifically designed for HSIC. First, we design a composite scanning Mamba block (CSMB) using Z-diagonal/horizontal traversals to model complex spatial–spectral interactions. Second, we develop a spatial–spectral masked Mamba (SAEM) that enforces robust feature learning via joint masking. Third, we integrate CL with multiview spectral splitting to enhance instance discrimination, creating a unified architecture that synergizes local reconstruction fidelity with global representation consistency.

The main contributions of this article are as follows.

1) We propose a CSMB with bidirectional Z-shaped and horizontal scanning paths, enabling simultaneous capture of local textures and global patterns for enhanced irregular object recognition in HSI.
2) We develop an SAEM featuring dual-branch masked reconstruction. By randomly masking 75% of spectral–spatial cubes, it learns robust representations of occluded features through cross-dimensional recovery.
3) We first pioneer the integration of MIM and CL in a Mamba architecture, where partitioned bands serve as augmented views to boost interclass discriminability while preserving intraclass consistency.
4) Extensive experiments demonstrate that our SSupMamba achieves state-of-the-art performance on four benchmarks with only five samples per class, proving its effectiveness for few-shot HSI classification.

The remainder of this article is organized as follows: Section II reviews technical foundations, Section III details SSupMamba's architecture, Section IV presents experimental validation, and Section V concludes with future directions.

II. Related Work

A. SSM and Mamba

SSMs have emerged as competitive sequence modeling frameworks, using differential equations to capture system dynamics through hidden state evolution [31]. The structured SSM (S4) variant initially addressed long-range dependency modeling but suffered from context-agnostic parameterization [21]. This limitation was overcome by Mamba (S6) [19], which introduced input-dependent selection mechanisms to dynamically filter sequence information. Its hardware-aware implementation achieved 3.8× throughput compared with transformers in language tasks, catalyzing vision adaptations. Zhu et al. [20] proposed vision Mamba (ViM) by flattening image patches into 1-D sequences, while Zhang et al. [32] introduced VMamba with a cross-scan module (CSM) that uses four-directional scanning to establish global receptive fields. In hyperspectral analysis, recent Mamba adaptations demonstrate progressive innovation: Li et al. [25] propose MambaHSI with adaptive spatial–spectral integration, while S2Mamba [33] uses composite horizontal–vertical spatial scanning and bidirectional spectral scanning, fused via a hybrid gating mechanism. Meanwhile, DualMamba [34] combines convolutional feature extraction with lightweight spatial and spectral Mamba branches, achieving cross-attention-based fusion for enhanced classification performance. However, these supervised approaches exhibit two critical limitations: 1) they require substantial labeled data that contradict practical HSIC scenarios where annotations are scarce and 2) their fixed scanning patterns (horizontal/vertical) fail to adequately model irregular geospatial features prevalent in HSI data. Our CSMB introduces a joint Z-diagonal and horizontal scanning to preserve irregular object continuity while maintaining linear complexity. Furthermore, we pioneer a self-supervised Mamba training through spatial–spectral masking, enabling effective few-shot adaptation without dense annotations.

B. MIM for HSIC

MIM has emerged as a powerful paradigm for self-supervised representation learning, where models predict randomly occluded image regions to develop robust features [35], [36]. Recent advancements in MIM for HSIC demonstrate diverse architectural strategies: Ibañez et al. [27] propose MAEST, a pixelwise spectral masking autoencoder that embeds spectral characteristics into transformers for classification, while Liu et al. [37] use PCA-based dimensionality reduction before spectral-dimension masked pretraining with transformers. Huang et al. [38] introduce spectral information aggregation (SIA) for dimension reduction combined with spatial masking, whereas SS-MAE [39] innovates through dual spatial–spectral autoencoders built on hybrid transformers to separately enhance spatial and spectral representations. These works collectively validate MIM's effectiveness in boosting HSIC performance through tailored masking mechanisms and self-supervised learning paradigms. Nevertheless, existing MIM-based HSIC methods face limitations in few-shot scenarios due to transformers' quadratic computational complexity and constrained spatial–spectral joint representation capabilities. To address this, we propose a CSMB backbone for efficient HSI patch modeling and SAEM, a self-supervised SAEM framework that enhances feature learning through joint reconstruction tasks, achieving superior representation and generalization with shallow encoders while maintaining linear computational efficiency.

C. CL for HSIC

Self-supervised CL, which learns representations by maximizing similarity between related samples and minimizing irrelevant ones without annotations, has shown remarkable success in few-shot natural image classification [40]. Following this success, HSIC researchers have developed domain-specific contrastive approaches: Liu et al. [41] proposed a deep multiview contrastive network (DMVL) using PCA-generated multiview samples for unsupervised training, while Guan and Lam [29] implemented domain-specific CL

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between spatial and spectral features to enhance classification performance. Recent explorations in hybrid frameworks combine MIM with CL, as seen in natural image studies [36] and HSIC attempts like C-SS-MTr [38], which integrates MIM reconstruction loss into CL branches. However, such hybrid approaches yield only marginal improvements compared with pure CL methods, particularly under low-sample conditions where performance remains suboptimal. This highlights the need for more effective integration strategies between self-supervised paradigms to address HSIC's unique spatial–spectral representation challenges in data-scarce scenarios. To address this, we propose SSupMamba, a novel network that synergistically integrates MIM and CL through dual SAEM modules extracting local spatial–spectral features from multiview HSI patches, coupled with a scalable synchronous contrastive framework for discriminative representation learning.

III. Methodology

This section begins with a brief introduction to Mamba fundamentals and an overview of the proposed SSupMamba framework. We then detail the core components: the CSMB and the SAEM. Finally, we describe the loss function and the fine-tuning strategy for classification.

A. Preliminaries

SSMs originate from the classical Kalman filter [42] and serve as a mathematical framework for describing the behavior of dynamic systems. SSM maps a 1-D input sequence $x(t) \in \mathbb{R}^L$ to an N-dimensional hidden state $h(t) \in \mathbb{R}^N$, which is eventually projected onto a 1-D output signal $y(t) \in \mathbb{R}^L$. The basic form of SSM is defined by the following equations:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t) \end{aligned} \tag{1}$$

Here, $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state matrix that determines the internal dynamics between states, $\mathbf{B} \in \mathbb{R}^{N\times L}$ is the input matrix, and $\mathbf{C} \in \mathbb{R}^{L\times N}$ and $\mathbf{D} \in \mathbb{R}^{L\times L}$ are the output matrix and the direct transmission matrix, respectively. $h'(t) \in \mathbb{R}^N$ represents the time derivative of the state $h(t) \in \mathbb{R}^N$. To address the significant challenges associated with integrating continuous systems into deep learning algorithms, structured SSMs (S4) and Mamba discretize continuous systems. Specifically, they use a time scale parameter $\Delta \in \mathbb{R}^L$ and apply the zero-order hold (ZOH) method to convert the continuous parameters $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ into discrete parameters $\overline{\boldsymbol{A}} \in \mathbb{R}^{N\times N}$ and $\overline{\boldsymbol{B}} \in \mathbb{R}^{N\times L}$. The specific definitions are as follows:

$$\begin{aligned} \overline{\boldsymbol{A}} &= \exp(\Delta \boldsymbol{A}), \quad \overline{\boldsymbol{C}} = \boldsymbol{C} \\ \overline{\boldsymbol{B}} &= (\Delta \boldsymbol{A})^{-1}\left(\exp(\Delta \boldsymbol{A}) - \boldsymbol{I}\right) \cdot \Delta \boldsymbol{B} \end{aligned} \tag{2}$$

The continuous state-space equations for the discretized SSM are as follows:

$$\begin{aligned} h(t) &= \overline{\boldsymbol{A}}h(t-1) + \overline{\boldsymbol{B}}x(t) \\ y(t) &= \overline{\boldsymbol{C}}h(t) \end{aligned} \tag{3}$$

Here, $x(t)$ and $y(t)$ represent the discrete inputs and outputs, respectively, while $h(t-1)$ and $h(t)$ denote the discrete hidden states. More details about the principles of Mamba can be found in [19].

B. Overview of the Proposed Model

To address the dual challenges of annotation dependence and spectral–spatial feature modeling in few-shot hyperspectral classification, we propose SSupMamba—a novel self-supervised framework with two-stage optimization. The architecture uses multidirectional spatial–spectral contextual modeling through its core CSMB, operating within a synergistic training paradigm.

1) Self-Supervised Pretraining Stage (Fig. 1): From the original HSI cube $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, densely sampled image patches $\mathbf{P}_i \in \mathbb{R}^{h\times w\times C}$ undergo band-group partitioning into complementary multiview inputs $\mathbf{V}_1$ and $\mathbf{V}_2 \in \mathbb{R}^{h\times w\times (C/2)}$. These spectral-split views serve as natural positive pairs for CL. Each view is processed through twin SAEM networks with shared-parameter encoders but independent decoders. The processing pipeline comprises: a spatial–spectral masked module (SSMM) that jointly occludes 75% of spatial–spectral tokens, a CSMB encoder extracting contextual features from unmasked regions, a dedicated CSMB decoder reconstructing original inputs, and a pixelwise reconstruction loss optimizing feature fidelity. Crucially, encoder outputs feed into a projection head where multiview CL aligns same-patch representations while dispersing dissimilar features, establishing a geometrically discriminative latent space.

2) Supervised Fine-Tuning Stage (Fig. 2): The pretrained encoder is adapted for classification by removing the SSMM, decoders, and projection heads, and then integrating a compact classification module. During end-to-end fine-tuning with limited annotations, the cross-entropy loss is used to refine the representations specifically for land cover classification, while retaining transferable spatial–spectral knowledge. Both fine-tuning and inference are performed using unmasked patches from the same single view (consistently either $\mathbf{V}_1$ or $\mathbf{V}_2$).

This dual-phase design delivers critical advantages. Self-supervision leverages unlabeled data to build transferable representations through occlusion-based reconstruction and spectral-view consistency. Subsequent fine-tuning enables data-efficient adaptation to downstream tasks, collectively overcoming annotation scarcity while modeling complex hyperspectral signatures.

C. Composite Scanning Mamba Block

While transformers [43] leverage self-attention to compute pairwise correlations among all the elements in an input sequence, enabling long-range dependency modeling and demonstrating widespread application in HSIC [13], [44], [45], their computational complexity scales quadratically with sequence length, incurring significant inefficiency. Mamba, founded on the selective SSM, establishes long-range dependencies among HSI spatial patches through linear-complexity state transitions, offering superior efficiency. However, the complex, irregular geometric structures inherent in HSI ground

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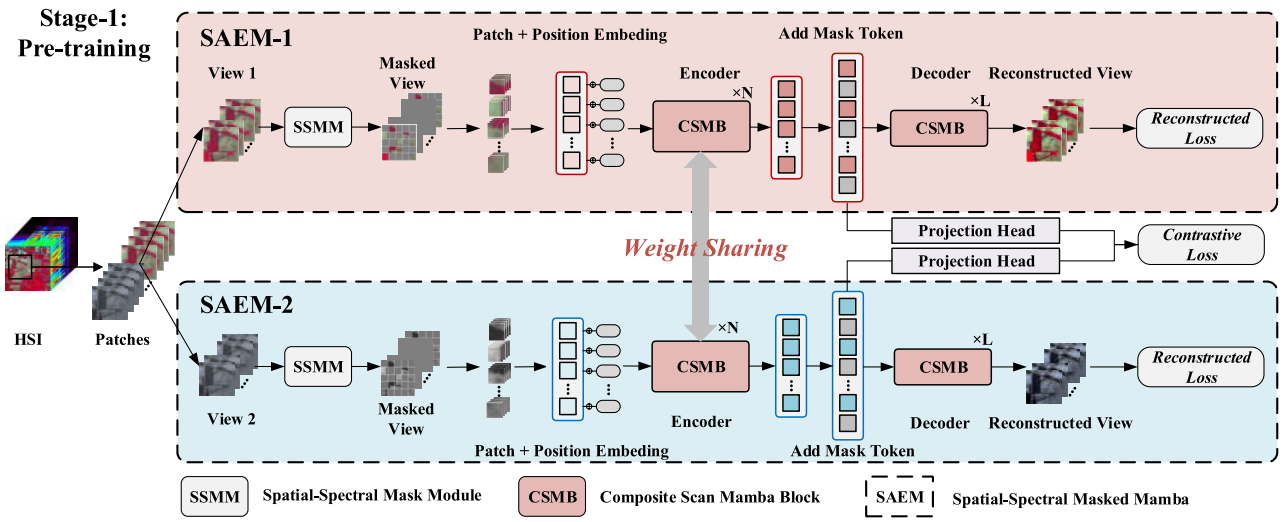


Fig. 1. Overview of the proposed SSupMamba. Given an HSI, samples are first collected centered around each ground object. The sampled patches are then divided into two groups on average by band, with each group being treated as a view of the HSI. During the self-supervised pretraining phase (Stage-1), these views are fed into two SAEM networks (SAEM-1/2). While architecturally identical, SAEM-1 and SAEM-2 specialize in learning view-specific representations through shared-encoder/independent-decoder configurations.

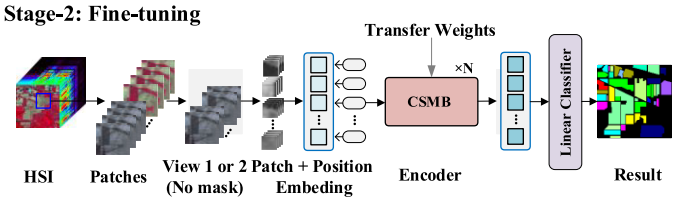


Fig. 2. Fine-tuning stage of the proposed SSupMamba. After pretraining, the encoding component of SAEM is used as a feature extractor, and a classification head is added to obtain the final results (Stage-2).

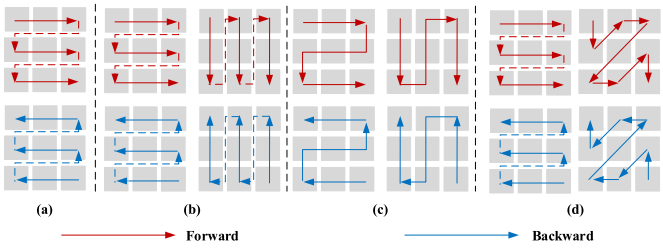


Fig. 3. Illustration of different scanning methods. (a) Horizontal. (b) Horizontal/vertical. (c) Horizontal/vertical zigzag. (d) Horizontal/diagonal zigzag.

objects—such as the sinuous boundaries of lakes or angular agricultural fields—pose a significant challenge. Standard scanning strategies, particularly simple flattening approaches, fundamentally prioritize Cartesian grid relationships (e.g., horizontal or vertical adjacencies). Consequently, they struggle to effectively capture the crucial diagonal spatial dependencies and off-axis spatial continuity that often define object morphology and class boundaries in complex scenes.

To address this critical limitation and significantly enhance spatial feature representation, we propose a CSMB. Its core innovation lies in using complementary scanning patterns: horizontal scanning (HScan) and zigzag diagonal scanning (ZScan), as illustrated in Fig. 3. Critically, the ZScan pattern traverses pixels along 45° diagonals in a continuous path. This angular orientation is intrinsically aligned with complex natural structures, explicitly capturing diagonal spatial relationships and preserving local coherence along relevant geometric directions—capabilities inherently lacking in axis-aligned scans. By integrating ZScan alongside HScan, the CSMB comprehensively models both the orthogonal and diagonal dependencies, capturing richer, geometrically invariant spatial contexts essential for classifying intricate land cover patterns.

As depicted in Fig. 4, the CSMB processes an input embedded sequence $\mathbf{L}_t \in \mathbb{R}^{B\times N\times D}$ (where B, N, and D denote batch size, sequence length, and feature dimension, respectively) for the tth block as follows: First, the input sequence passes through a normalization layer.

$$\mathbf{L}_t' = \text{Norm}\left(\mathbf{L}_t\right) \tag{4}$$

Then, the sequence is unfolded through ZScan and HScan, with bidirectional processing applied, and linear layers are used to obtain s and z, respectively.

$$\mathbf{s}_1, \mathbf{s}_2 = \text{Linear}\left(\text{ZS}_{\text{forward}}\left(\mathbf{L}_t'\right), \text{ZS}_{\text{backward}}\left(\mathbf{L}_t'\right)\right) \tag{5}$$

$$\mathbf{s}_3, \mathbf{s}_4 = \text{Linear}\left(\text{HS}_{\text{forward}}\left(\mathbf{L}_t'\right), \text{HS}_{\text{backward}}\left(\mathbf{L}_t'\right)\right) \tag{6}$$

$$\mathbf{z} = \text{Linear}\left(\mathbf{L}_t'\right) \tag{7}$$

For each direction of s, 1-D convolution, SiLU activation, and SSM are applied separately.

$$\mathbf{s}_1', \mathbf{s}_2' = \text{SSM}\left(\text{SiLU}\left(\text{Conv1}d\left(\mathbf{s}_1, \mathbf{s}_2\right)\right)\right) \tag{8}$$

$$\mathbf{s}_3', \mathbf{s}_4' = \text{SSM}\left(\text{SiLU}\left(\text{Conv1}d\left(\mathbf{s}_3, \mathbf{s}_4\right)\right)\right) \tag{9}$$

$$\mathbf{z}' = \text{SiLU}\left(\mathbf{z}\right) \tag{10}$$

Finally, the residual connection produces the final output

$$\mathbf{L}_{t+1} = \text{Linear}\left(\mathbf{z}' \odot \left(\text{Ra}\left(\mathbf{s}_1' + \mathbf{s}_2'\right) + \mathbf{s}_3' + \mathbf{s}_4'\right)\right) + \mathbf{L}_t \tag{11}$$

Here, Ra(∗) denotes the sequence backtracking operation, and $\mathbf{L}_{t+1}$ is the final output of the tth block.

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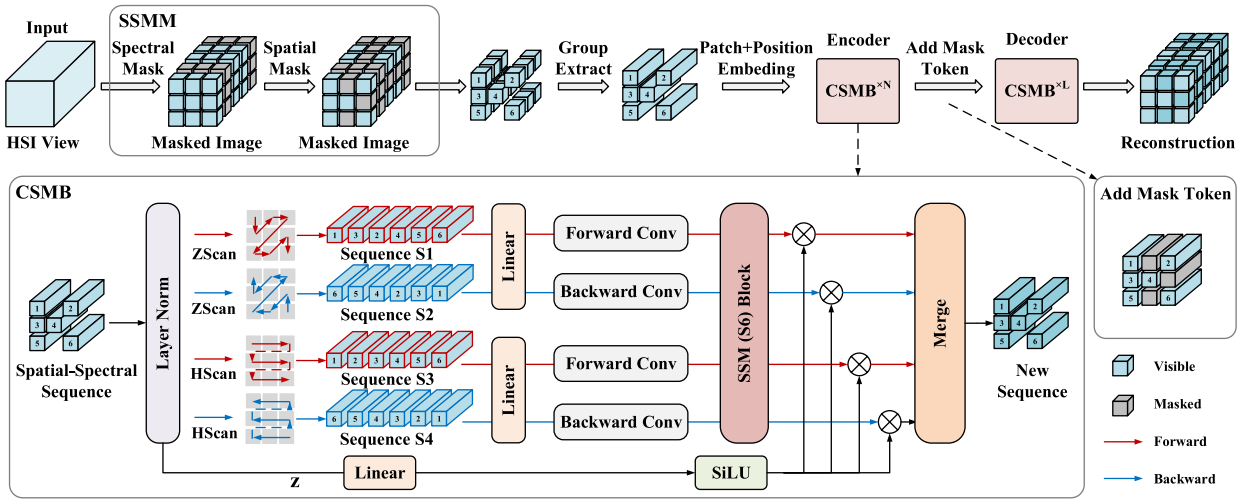


Fig. 4. SAEM and SSMM. Given image blocks from a view, they are cropped according to a specific neighborhood size and then input into SSMM. Subsequently, spectral band masking and spatial masking are applied sequentially based on specific masking rates. The CSMB—the spatial–spectral embedding sequences of HSI—traverse along four different scanning paths, each sequence independently processed by different S6 blocks. Subsequently, the results are merged into a feature map as the ultimate output (cross-merging).

D. Spatial–Spectral Masked Mamba

MIM reduces annotation dependence through self-supervised pretraining and enhances feature extraction via pixel-level reconstruction—attributes particularly suited to HSI given its high-dimensional characteristics. While effective, transformer-based MIM implementations incur prohibitive computational complexity for HSI processing. To address this while specifically targeting few-shot learning (FSL) limitations, we propose an SAEM framework (Fig. 4). SAEM integrates: 1) a dual-stage SSMM applied directly to fully spectral HSI data and 2) an asymmetric autoencoder architecture built upon our efficient CSMBs in Section III-C. Critically distinct from prior work that often masks dimensionality-reduced features, our SSMM module operates directly on the input spectral domain of the original HSI patch prior to spatial tokenization. This approach preserves the intrinsic richness of the raw spectral data during masking, enhancing representation learning. This synergistic design significantly enhances discriminative feature representation under scarce annotations while maintaining linear computational complexity.

The SSMM processes each view through hierarchical masking, preserving input spectral fidelity.

Spectral Masking: Applied directly to the spectral bands of $\mathbf{V}_i$, a fixed ratio r_{Spe} retains $C_{\text{BM}} = (1 - r_{\text{Spe}}) \times C/2$ spectral bands, generating spectrally sparse inputs $\mathbf{V}'_i \in \mathbb{R}^{h \times w \times C_{\text{BM}}}$—a masking operation performed on the native spectral dimension.

Spatial Tokenization and Masking: $\mathbf{V}'_i$ is partitioned into $N = hw/P^2$ nonoverlapping patches $\{\mathbf{x}_p^j\}$ of size $P \times P$. Each patch is linearly projected using a weight matrix $\mathbf{E} \in \mathbb{R}^{P^2 \times C_{\text{BM}} \times D}$ and augmented with positional encodings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{N \times D}$:

$$\mathbf{Z}_0 = \left[\mathbf{x}_p^1\mathbf{E}; \cdots; \mathbf{x}_p^N\mathbf{E}\right] + \mathbf{E}_{\text{pos}} \tag{12}$$

Spatial masking (PM) then removes $r_{\text{Spa}} \times N$ patches from this tokenized representation, preserving $N_{\text{PM}} = (1 - r_{\text{Spa}}) \times N$ tokens

$$\mathbf{V}''_i = \text{PM}\left(\mathbf{Z}_0\right) \in \mathbb{R}^{N_{\text{PM}} \times D} \tag{13}$$

The remaining tokens are subsequently grouped according to their spatial positions and traversed along the spectral direction to form spatial–spectral sequences, which are then fed into the CSMB module.

Crucially, SAEM implements a reconstruction-oriented processing flow: The masked representation $\mathbf{V}''_i$ (appended with a class token) is encoded by four cascaded CSMBs. Before decoding, the full spectral–spatial sequence is reconstituted by reinserting masked spectral bands as zero-filled vectors and spatial tokens as masked positions. This complete sequence is reconstructed by a decoder comprising three CSMBs—a deliberate asymmetry optimizing abstraction–reconstruction balance. The reconstruction loss is

$$\mathcal{L}_{\text{mse}} = \frac{1}{n}\sum_{i=1}^{n}\left(\tilde{\mathbf{Y}}_i - \mathbf{Y}_i\right)^2 \tag{14}$$

where $\mathbf{Y}_i$ represents the ground truth, and $\tilde{\mathbf{Y}}_i$ represents the predicted value. It forces the network to recover fine-grained spatial–spectral signatures from limited visible context, leveraging CSMB's capacity for directional feature aggregation (Section III-C). By masking the raw HSI spectral bands directly—rather than latent features—SAEM more effectively captures intrinsic spectral–spatial dependencies crucial for few-shot generalization.

E. Self-Supervised Mamba Network

To further enhance the discriminative power of the SAEM, we integrate CL into a unified self-supervised Mamba network (SSupMamba). Our approach exploits a fundamental property of hyperspectral imagery: while individual spectral bands capture distinct physical attributes of ground objects, their underlying semantic content remains consistent across the spectral dimension. Motivated by cross-modal contrastive

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frameworks [40], [41], we treat each spectral band as a unique view of the same semantic scene. This intrinsic spectral diversity provides natural positive sample pairs for self-supervised representation learning, enabling discriminative feature acquisition without manual annotations. We implement a scalable batch-view generation strategy to efficiently construct these spectral-view pairs.

As illustrated in Fig. 1, SSupMamba extends SAEM with a CL branch. The SSMM applies dual-stage masking to input views. A shared encoder processes masked inputs while two separate decoders reconstruct the original spectral–spatial content for each view. A projection head, comprising two fully connected (linear) layers with intermediate batch normalization (BN) and ReLU activation, maps the encoded features to the CL space. The contrastive loss operates on the projected embeddings $\mathbf{Z}_i$, $\mathbf{Z}_j$ from paired spectral views:

$$\mathcal{L}_{\mathrm{CL}} = -\log \frac{\exp\left(\mathrm{sim}\left(\mathbf{Z}_i, \mathbf{Z}_j\right)\right)}{\sum_{k=1}^{2N} l_{[k\neq i]} \exp\left(\mathrm{sim}\left(\mathbf{Z}_i, \mathbf{Z}_k\right)\right)} \tag{15}$$

where 2N represents total views per batch (N samples $\times$ 2 spectral views), $l_{[k\neq i]}$ represents indicator function excluding positive pairs, and sim(·) represents cosine similarity.

This objective clusters embeddings from the same spatial location across spectral bands (positive pairs) while repelling those from different locations (negative pairs). The total loss combines reconstruction and contrastive objectives:

$$\mathcal{L}_{\mathrm{SSupMamba}} = \mathcal{L}_{\mathrm{mse}_{V1}} + \mathcal{L}_{\mathrm{mse}_{V2}} + \lambda \mathcal{L}_{\mathrm{CL}} \tag{16}$$

Here, $\mathcal{L}_{\mathrm{mse}_{V1}}$ and $\mathcal{L}_{\mathrm{mse}_{V2}}$ denote reconstruction losses for the two spectral views, while λ balances the contribution of the contrastive objective. This multitask optimization enables SSupMamba to learn both semantically coherent embeddings and spectrally rich representations without manual annotations. After pretraining, the encoder is transferred to downstream classification tasks by replacing the projection head with a classifier while retaining its powerful feature extraction capabilities.

IV. Experiments

This section provides a comprehensive evaluation of the proposed framework. We begin by introducing the four benchmark hyperspectral datasets used for validation, along with key experimental settings. We then compare our method against state-of-the-art HSIC approaches, perform ablation studies to validate core components, visualize the learned features, and analyze computational efficiency.

A. Dataset Description

Experiments were conducted on four benchmark HSI datasets representing diverse acquisition scenarios and land cover complexities:

1) WHU-Hi-LongKou (LK): Acquired via Headwall Nano-Hyperspec sensor, this dataset covers an agricultural area comprising 550 $\times$ 400 pixels with 2 04 542 annotated pixels distributed across 16 land cover classes.

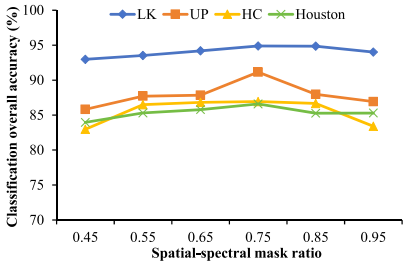


Fig. 5. OA(%) with different spatial–spectral mask ratios on four HSI datasets.

2) University of Pavia (UP): Captured by the ROSIS airborne sensor, this urban scene contains 103 spectral bands with 42 776 labeled pixels and nine distinct land-use categories.

3) WHU-Hi-HanChuan (HC): Collected over Hanchuan, China, this large-scale scene (1217 $\times$ 303 pixels) includes 274 spectral bands spanning visible to near-infrared ranges and 16 land cover types.

4) Houston2013 (HU): Acquired by the National Center for Airborne Laser Mapping for the 2013 IEEE GRSS Data Fusion Contest, this dataset (349 $\times$ 1905 pixels, 144 bands) contains 15 029 labeled pixels across 15 urban land cover classes.

B. Experimental Configuration

All the experiments were implemented in PyTorch and executed on a workstation with an Intel i9-13900KF CPU and NVIDIA RTX 4090 GPU. We used the Adam optimizer [46] with a learning rate of $1e^{-3}$ and weight decay of $1e^{-6}$. Input patches were configured as 28 $\times$ 28 pixel neighborhoods with a batch size of 128. During self-supervised pretraining, we applied 75% spatial–spectral masking across 300 epochs, followed by 180 fine-tuning epochs. Performance was evaluated using standard metrics: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ), providing complementary perspectives on classification fidelity.

C. Parameter Sensitivity Analysis

We systematically investigate the impact of two critical hyperparameters on SSupMamba's performance: spatial–spectral masking rate and neighborhood size.

1) Mask Rate: Masking ratios were evaluated across 45%–95% (10% increments) as shown in Fig. 5. Performance consistently improved with increasing masking rates up to 75% across all the datasets, demonstrating the importance of challenging reconstruction tasks for representation learning. Beyond 75%, accuracy markedly declined—excessive masking critically compromises feature recovery capability. This establishes 75% as the optimal threshold balancing sufficient signal preservation with feature abstraction difficulty.

2) Neighborhood Size: Neighborhood size was tested from 22 $\times$ 22 to 32 $\times$ 32 pixels (Fig. 6). Smaller neighborhoods (22–26) showed degraded performance due to inadequate spatial context. Peak accuracy occurred at 28 for the LK, HC, and HU datasets, while UP achieved optimal results at 26. Since 28 yielded maximal performance for three datasets and near-optimal results for UP, this size was standardized for architectural consistency and computational efficiency.

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TABLE I

CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE WHU-HI-LONGKOU DATASET

Type	Supervised Learning			Few-shot Learning	Self-supervised Learning				
Class	PCA+SVM	SSTN[44]	MambaHSI[25]	DCFSL[47]	SS-MTr[38]	C-SS-MTr[38]	TCRL[48]	CMLPL[49]	SSupMamba
1	57.42±0.77	97.16±1.21	98.05±1.08	**99.67±1.31**	95.35±1.18	93.36±1.26	97.71±0.99	99.14±0.76	98.56±1.86
2	74.19±0.75	57.32±0.46	78.61±0.62	89.79±0.64	83.69±1.17	85.92±1.12	**98.52±0.98**	93.76±0.64	95.42±1.36
3	44.17±0.66	86.84±1.22	94.69±1.56	92.14±1.47	89.34±0.55	88.89±0.72	95.87±1.31	95.06±0.65	**98.94±1.22**
4	41.43±0.74	88.39±0.83	55.61±1.03	90.17±0.27	83.14±0.47	82.56±0.94	90.11±0.74	88.22±0.41	**91.89±1.23**
5	69.59±0.93	78.11±0.54	89.39±0.42	94.31±1.08	77.75±1.13	79.75±1.17	92.62±0.59	65.05±0.51	**98.87±1.45**
6	67.69±1.19	77.46±1.07	97.57±1.20	89.84±0.45	88.40±0.73	88.66±1.33	**98.86±0.51**	90.39±0.79	97.77±1.32
7	83.28±0.78	**99.98±0.93**	99.79±0.94	99.97±1.16	92.08±1.01	92.79±1.31	94.20±1.15	99.37±0.94	97.57±1.38
8	54.31±1.03	65.77±0.35	**87.64±0.68**	71.50±0.67	57.36±1.18	66.00±1.10	70.97±1.19	71.81±0.44	74.09±2.15
9	41.00±0.58	45.65±0.38	78.27±0.36	81.72±0.32	81.35±0.64	80.47±1.26	82.66±0.72	60.83±0.34	**88.10±1.29**
OA(%)	58.46±1.49	89.58±0.46	83.60±0.77	89.89±0.71	87.74±2.10	89.27±0.96	90.07±1.30	91.45±0.63	**94.89±1.49**
AA(%)	58.19±1.55	77.41±1.31	86.63±1.13	88.90±1.25	83.25±1.45	86.24±1.10	89.41±0.68	83.37±0.36	**93.47±1.04**
k×100	50.68±0.69	86.42±1.22	85.00±0.68	88.26±0.82	84.32±0.90	84.11±0.69	88.27±0.89	88.86±1.22	**93.37±1.90**

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE UNIVERSITY OF PAVIA DATASET

Type	Supervised Learning			Few-shot Learning	Self-supervised Learning				
Class	PCA+SVM	SSTN[44]	MambaHSI[25]	DCFSL[47]	SS-MTr[38]	C-SS-MTr[38]	TCRL[48]	CMLPL[49]	SSupMamba
1	39.98±0.02	65.79±3.01	84.22±1.24	76.94±0.07	66.36±1.48	71.29±1.42	83.19±0.67	73.19±0.99	**85.99±1.03**
2	96.51±2.01	33.20±4.20	82.89±1.35	71.77±1.08	71.66±1.44	70.31±0.98	73.30±1.03	88.83±0.68	**97.50±0.60**
3	39.17±0.24	48.69±2.62	65.07±2.15	51.19±1.14	72.78±0.90	75.46±1.03	83.56±0.63	80.65±1.26	**92.85±1.80**
4	41.78±1.72	94.36±0.86	83.60±1.82	**96.70±0.42**	63.42±0.70	59.74±1.70	76.97±0.59	90.23±1.17	69.06±0.73
5	40.31±0.97	**100.0±0.00**	99.47±0.04	98.98±0.02	98.35±1.43	99.20±1.24	98.91±1.61	**100.0±0.00**	**100.0±0.00**
6	38.92±1.30	98.10±0.50	95.93±0.69	75.67±0.95	82.08±0.71	83.52±1.44	92.58±0.72	87.22±1.38	**98.67±1.27**
7	40.80±0.28	**99.84±1.61**	69.35±1.36	76.14±1.61	91.56±1.58	90.41±1.68	97.31±0.80	97.47±1.51	95.66±1.55
8	41.28±1.01	96.03±0.30	87.35±2.10	61.17±1.16	86.31±0.77	84.75±1.59	96.31±0.58	71.77±0.65	**96.44±1.52**
9	40.51±0.69	96.86±0.22	98.18±0.43	**98.94±1.33**	54.50±1.08	54.77±1.99	72.79±0.76	87.66±0.44	48.99±0.95
OA(%)	66.75±0.77	61.67±2.39	84.63±0.83	74.90±0.13	73.86±0.70	73.94±3.80	81.30±1.26	87.42±1.18	**91.16±1.64**
AA(%)	46.80±0.12	81.43±1.06	85.12±1.04	78.61±0.23	76.34±0.86	76.61±2.54	85.10±1.43	85.14±0.55	**86.08±0.78**
k×100	47.71±0.69	55.11±2.48	82.50±1.55	68.58±0.58	67.18±0.95	67.28±1.31	76.60±0.96	83.67±0.93	**88.29±1.02**

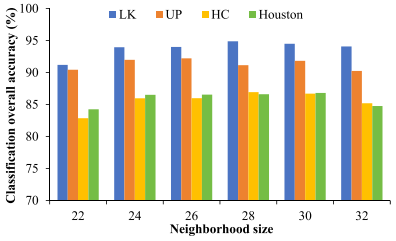


Fig. 6. OA(%) with different neighborhood sizes on four HSI datasets.

Final hyperparameters were fixed at 75% masking rate and 28 × 28 neighborhood size for subsequent experiments to ensure optimal performance across diverse hyperspectral domains.

D. Comparison to the State-of-the-Art Methods

We evaluate the proposed SSupMamba against eight representative HSIC methods covering diverse paradigms: the conventional spectral-feature approach PCA+SVM; state-of-the-art supervised deep learning models including the spatial–spectral transformer network SSTN [44] and the Mamba-based model MambaHSI [25]; the advanced FSL method DCFSL [47]; and contemporary self-supervised techniques, including masked transformer variants SS-MTr [38] and C-SS-MTr [38], the triplet contrastive method TCRL [48], and the contrastive semisupervised framework CMLPL [49]. For fair comparison under few-shot settings, each method uses the same five randomly selected samples per class per trial—without labels during pretraining and with labels during fine-tuning (or training). DCFSL is an exception, using its original pretraining dataset. The results are averaged over ten independent trials. Hyperparameters strictly follow the original publications: PCA+SVM retains 70 principal components, with SVM kernel optimization via fourfold cross-validation.

1) Quantitative Analysis: Quantitative results comparing different methods are presented in Tables I–IV, which include OA, AA, κ, per-class accuracy, and variance to thoroughly evaluate performance across datasets. The results clearly show the superiority of deep supervised learning methods (SSTN, MambaHSI) over traditional PCA+SVM across all the datasets, owing to their ability to learn hierarchical features and model complex nonlinear relationships. However, these supervised methods exhibit unstable performance under few-shot conditions. For example, on the LK dataset, class 7 achieves high accuracy, while classes 3 and 4 perform poorly. Similar inconsistencies are observed for class 3 in UP, classes 2, 8, and 9 in HC, and classes 10 and 12 in HU. This reflects the difficulty of modeling complex distributions with only five samples per class, often leading to overfitting. Although MambaHSI shows slightly improved stability, its overall performance remains limited. In contrast, methods leveraging unlabeled data—such as DCFSL, SS-MTr, C-SS-MTr, TCRL,

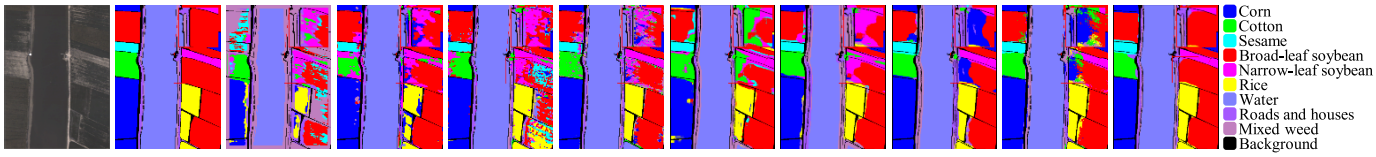
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TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE HU-HI-HANCHUAN DATASET

Type	Supervised Learning			Few-shot Learning	Self-supervised Learning				
Class	PCA+SVM	SSTN[44]	MambaHSI[25]	DCFSL[47]	SS-MTr[38]	C-SS-MTr[38]	TCRL[48]	CMLPL[49]	SSupMamba
1	41.61±1.48	72.22±0.63	87.11±1.31	85.99±1.41	66.37±0.73	67.47±1.49	80.13±0.98	**95.23±0.95**	90.64±1.57
2	54.63±1.25	35.01±0.75	58.67±2.01	34.56±0.82	55.48±1.35	56.27±0.77	71.02±1.39	45.38±1.06	**77.96±0.68**
3	50.33±0.87	74.39±0.66	74.53±0.65	55.40±1.20	53.48±1.09	57.82±1.31	73.97±1.18	64.10±1.15	**90.77±1.33**
4	79.95±1.32	74.46±0.79	90.69±1.14	89.60±1.31	80.76±1.08	83.48±1.33	95.78±1.35	97.15±1.16	**98.14±1.13**
5	97.16±1.38	73.22±0.85	**98.99±0.80**	96.32±0.91	88.24±1.28	93.02±0.97	97.45±0.81	93.64±0.67	98.94±0.78
6	17.31±1.43	**70.90±1.15**	38.36±2.13	21.86±1.28	49.12±1.15	49.13±1.21	58.99±1.48	47.87±0.59	70.30±1.22
7	87.61±0.55	95.66±1.27	**96.40±0.74**	79.52±0.63	74.17±1.26	73.71±1.46	83.72±1.13	**96.40±0.61**	91.87±0.79
8	20.68±1.41	55.71±0.76	28.60±1.38	30.56±0.72	39.36±0.51	48.25±0.92	58.75±0.75	74.74±0.52	**75.83±0.74**
9	12.15±0.97	36.33±0.66	30.70±0.55	33.90±1.47	45.75±1.06	48.01±1.34	60.25±1.03	46.36±0.73	**65.60±0.77**
10	18.57±0.78	74.69±0.95	87.45±0.41	78.41±1.05	52.17±0.90	51.57±1.16	79.85±1.30	**93.55±1.29**	92.84±0.80
11	70.42±0.71	41.99±0.65	66.87±2.05	66.13±0.77	62.17±1.30	63.06±0.87	76.55±1.36	80.41±0.37	**86.87±0.81**
12	82.76±1.46	47.25±1.25	88.18±1.01	38.65±1.36	67.61±0.50	68.65±1.49	83.44±1.14	87.61±0.78	**96.22±1.32**
13	30.37±1.23	39.31±1.43	46.23±1.20	53.45±0.46	38.97±0.90	40.98±1.32	50.72±0.68	44.69±1.28	**69.21±0.63**
14	48.92±1.43	49.89±0.58	52.88±0.83	56.02±0.39	47.15±1.02	53.08±0.51	55.01±0.74	48.32±0.87	**74.27±1.13**
15	81.60±1.24	**87.81±1.58**	80.73±0.49	55.97±0.52	82.77±1.48	81.68±1.35	87.25±1.37	87.08±0.53	87.57±0.60
16	60.78±0.76	97.97±1.53	91.66±0.85	77.11±1.21	85.38±0.67	86.61±0.81	85.78±0.95	**99.20±1.23**	95.83±0.77
OA(%)	49.45±0.74	65.73±1.06	73.51±1.02	64.95±1.17	64.96±1.08	67.00±1.22	75.61±1.80	78.33±1.31	**86.94±1.10**
AA(%)	46.67±1.14	58.22±0.77	69.81±0.95	59.59±0.83	61.80±1.28	63.92±0.86	74.91±1.13	74.80±0.98	**85.17±1.24**
k×100	43.13±0.50	60.51±0.67	73.33±1.14	60.04±1.08	59.92±1.25	62.30±0.50	72.02±1.23	74.74±1.26	**84.82±1.52**

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE HOUSTON DATASET

Type	Supervised Learning			Few-shot Learning	Self-supervised Learning				
Class	PCA+SVM	SSTN[44]	MambaHSI[25]	DCFSL[47]	SS-MTr[38]	C-SS-MTr[38]	TCRL[48]	CMLPL[49]	SSupMamba
1	25.89±2.94	85.78±0.59	94.34±0.69	**98.96±0.16**	77.96±1.07	68.34±2.85	78.45±0.82	79.84±0.56	87.02±0.66
2	38.51±1.18	70.42±1.86	70.30±1.23	83.43±1.18	65.89±0.51	55.64±2.92	68.02±0.77	68.94±1.15	**87.31±0.80**
3	69.58±3.47	94.97±1.97	**97.07±1.06**	93.93±0.74	94.89±1.21	91.01±1.97	94.34±1.44	95.76±1.43	95.80±1.06
4	61.49±4.28	84.31±1.41	**94.39±0.74**	84.67±1.87	55.18±1.29	50.54±0.54	77.79±0.90	72.25±0.80	86.75±0.51
5	25.70±1.96	**100.0±0.00**	99.92±0.05	98.46±2.00	96.29±0.81	91.56±0.35	95.33±1.16	98.28±0.86	99.80±0.39
6	21.61±1.19	78.68±0.02	**91.94±0.61**	83.75±1.56	86.85±1.13	87.53±0.01	86.43±0.71	88.12±1.36	88.98±1.33
7	46.64±2.01	45.51±2.02	61.69±0.58	73.16±2.38	79.20±0.36	67.07±0.72	74.52±1.10	78.08±0.70	**84.14±0.66**
8	29.74±3.88	28.92±1.06	33.44±1.91	39.71±0.10	58.83±0.64	48.28±2.16	49.70±1.21	55.59±0.81	**64.46±0.74**
9	67.09±2.43	81.25±1.16	74.13±0.79	82.12±1.17	71.84±0.76	67.81±1.30	77.96±0.63	72.61±0.92	**84.09±0.70**
10	35.61±0.66	32.80±0.24	63.86±1.42	55.73±1.36	87.23±1.16	73.28±1.27	84.24±0.61	83.07±1.20	**89.84±1.28**
11	52.79±1.78	35.72±2.78	76.64±1.37	48.46±1.17	82.87±0.92	65.22±0.54	76.82±0.81	72.50±0.93	**88.92±0.76**
12	42.90±4.28	8.16±0.82	56.40±1.33	71.42±1.97	70.54±1.05	55.38±0.18	64.73±1.34	65.75±1.45	**76.45±1.37**
13	23.42±1.22	**90.42±1.50**	86.56±0.36	89.22±0.41	86.88±0.93	81.07±1.05	80.62±0.85	83.83±1.26	90.10±1.45
14	30.65±0.10	**100.0±0.00**	**100.0±0.00**	81.80±2.01	99.97±0.71	97.30±0.60	99.59±1.20	**100.0±0.00**	**100.0±0.00**
15	21.21±0.05	**100.0±0.00**	99.07±1.52	96.64±0.14	97.36±1.37	95.67±1.19	94.44±1.26	96.99±0.70	97.55±1.44
OA(%)	46.45±2.49	62.33±1.07	76.41±1.28	76.55±0.26	77.86±0.64	68.87±3.28	77.65±1.09	78.93±1.24	**86.60±1.41**
AA(%)	39.48±1.76	69.13±0.01	80.98±1.01	78.76±0.36	80.79±1.15	73.05±2.50	80.20±0.80	80.77±0.65	**88.08±0.79**
k×100	41.69±2.95	59.74±1.05	81.43±0.95	74.66±0.31	76.10±0.83	66.40±3.50	75.86±1.33	77.17±1.39	**85.53±0.59**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 7. Classification maps for the WHU-Hi-LongKou dataset. (a) False-color image (29, 19, 9). (b) Ground truth. (c) PCA+SVM. (d) SSTN. (e) MambaHSI. (f) DCFSL. (g) SS-MTr. (h) C-SS-MTr. (i) TCRL. (j) CMLPL. (k) Proposed SSupMamba.

and CMLPL—demonstrate better robustness. The proposed SSupMamba outperforms all these, especially on previously challenging categories. For instance, on the LK dataset, it achieves 98.94% (class 3) and 91.89% (class 4), improving upon CMLPL by 3.99% and 3.67%, respectively. SSupMamba also achieves the highest OA, AA, and κ on all the four datasets. On the challenging HU dataset, it attains OA and AA of 86.60% and 88.08%, outperforming CMLPL by 7.67% and 7.31%, and exceeding C-SS-MTr by over 5%. These gains are attributed to SSupMamba's self-supervised masked pretraining strategy, which learns comprehensive spatial–spectral features. By integrating MIM with CL, it obtains highly discriminative representations that enable fast and accurate downstream classification.

2) Qualitative Evaluation: Figs. 7–10 present false-color composites, ground truth, and classification results from eight methods across four hyperspectral datasets. Visual comparison reveals notable differences in map quality. The

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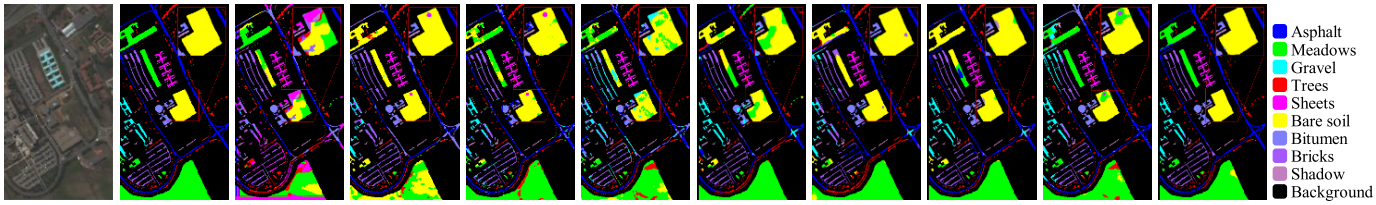


Fig. 8. Classification maps for the UP dataset. (a) False-color image (29, 19, 9). (b) Ground truth. (c) PCA+SVM. (d) SSTN. (e) MambaHSI. (f) DCFSL. (g) SS-MTr. (h) C-SS-MTr. (i) TCRL. (j) CMLPL. (k) Proposed SSupMamba.

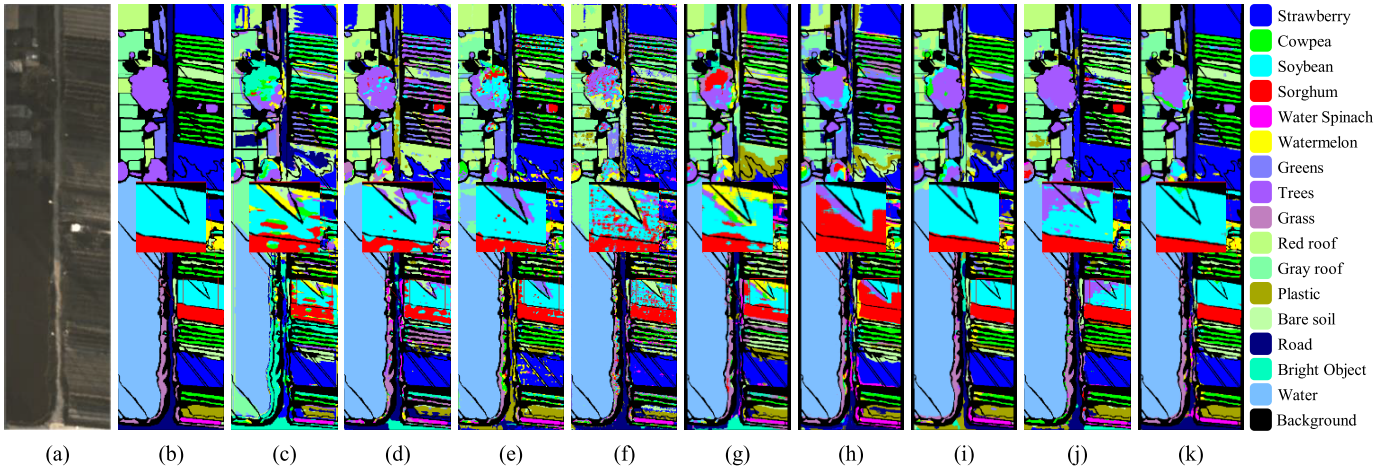


Fig. 9. Classification maps for the WHU-Hi-HanChuan dataset. (a) False-color image (29, 19, 9). (b) Ground truth. (c) PCA+SVM. (d) SSTN. (e) MambaHSI. (f) DCFSL. (g) SS-MTr. (h) C-SS-MTr. (i) TCRL. (j) CMLPL. (k) Proposed SSupMamba.

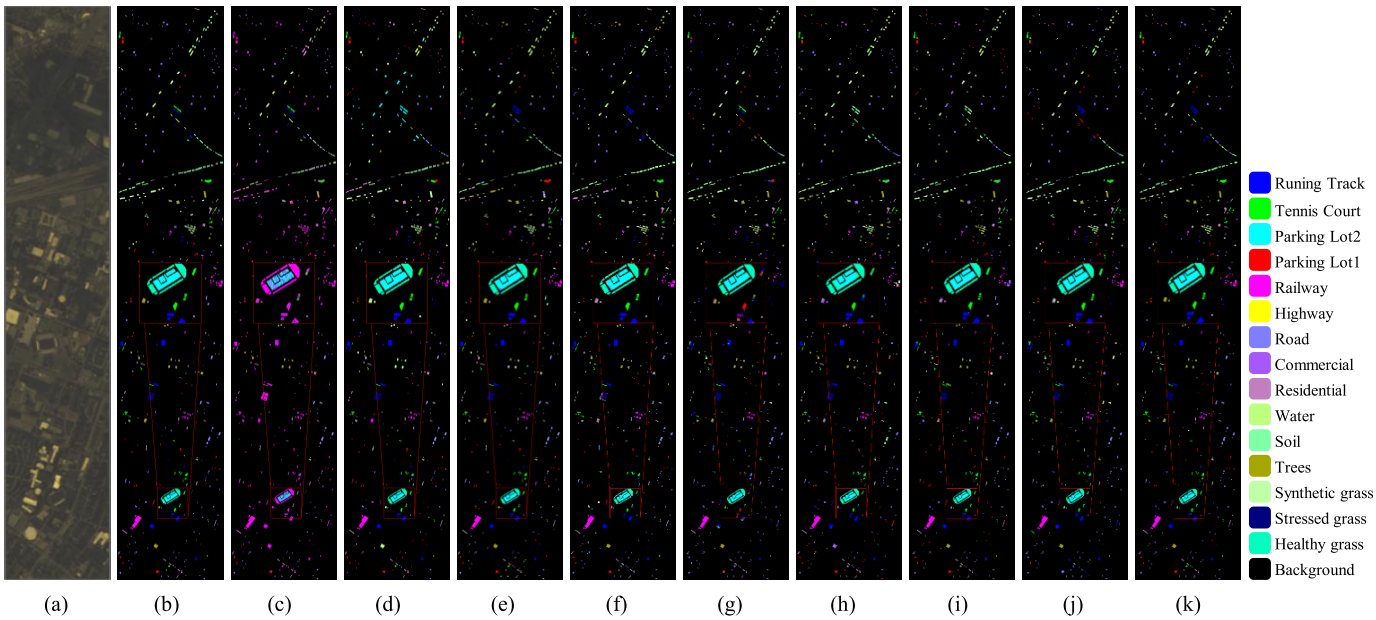


Fig. 10. Classification maps for the HU dataset. (a) False-color image (29, 19, 9). (b) Ground truth. (c) PCA+SVM. (d) SSTN. (e) MambaHSI. (f) DCFSL. (g) SS-MTr. (h) C-SS-MTr. (i) TCRL. (j) CMLPL. (k) Proposed SSupMamba.

conventional PCA+SVM approach produces noisy classifications with substantial artifacts, reflecting its limited capacity to model complex spectral–spatial relationships. Deep-learning-based methods effectively reduce large-area noise and yield improved spatial coherence. Consistent with quantitative results, supervised deep learning methods (SSTN, MambaHSI) exhibit clear interclass performance variations. While some classes are well-classified, others show pronounced misalignment—particularly evident in LK, UP, and HC—confirming the challenge of modeling complex distributions with only five labels per class. Methods incorporating unlabeled data—including transfer learning (DCFSL)

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TABLE V
OA(%) OF ABLATION STUDY ON SCANNING PATTERNS

Patterns	LK	UP	HC	HU
Horizontal	93.81±2.32	88.28±1.68	85.33±2.94	83.86±2.17
Horizontal+Vertical	93.27±1.47	88.39±1.82	84.75±3.35	82.91±2.06
H+V(Zigzag)	93.79±1.76	89.13±1.19	85.90±2.36	83.88±1.03
CSMB	**94.89±1.49**	**91.16±1.64**	**86.94±1.10**	**86.60±1.41**

and self-supervised approaches (SS-MTr, C-SS-MTr, TCRL, CMLPL)—partially alleviate these issues. DCFSL and C-SS-MTr, for instance, show fewer large misclassified regions, underscoring the value of unlabeled samples. However, scattered misclassifications persist. Visually, SSupMamba delivers superior fidelity across all the datasets. It markedly reduces both large and small misclassification patches, particularly in LK, UP, and HC. Notably, SSupMamba maintains consistently high accuracy across all the classes under limited samples, avoiding the uneven performance seen in other methods. This improvement arises from SSupMamba's pretraining design: MIM fosters robust spatial–spectral feature learning, while CL further refines discriminability. Their combination effectively resolves classification ambiguity, even for easily confusable pixels.

E. Ablation Study

1) Scanning Patterns: We evaluated the proposed CSMB against three established selective scanning patterns: horizontal (H), horizontal+vertical (H+V), and horizontal+vertical with zigzag processing (H+V(ZigZag)). Table V results show H+V(ZigZag) offered marginal improvement over H+V. Critically, CSMB's multidirectional scanning strategy significantly outperformed all the three patterns across all the datasets. This demonstrates that capturing oblique spatial–spectral information is essential for modeling complex HSI structures under limited samples, validating CSMB's design.

2) Masking Strategies: The efficacy of the SSMM was tested by comparing it to SAEM (V1) configurations using only spectral masking (SpeMask) or only spatial masking (SpaMask). The results (Table VI) reveal that SpeMask generally yielded better performance than SpaMask on the UP, LK, and HU datasets, indicating spectral features provide strong inherent discriminative power. Most importantly, the integrated SSMM strategy consistently surpassed both single-mask approaches across all the datasets. This superiority confirms that combining spectral and spatial masking creates a sufficiently challenging pretext task, forcing the model to learn comprehensive spatial–spectral representations crucial for downstream classification.

3) Learning Paradigm: The synergy between MIM and CL was analyzed by comparing SSupMamba (SAEM+CL) against SAEM variants relying solely on masked modeling. The results (Table VI) consistently show the addition of CL significantly enhances classification performance beyond masked modeling alone. This improvement stems from complementary learning objectives: SAEM focuses on reconstructing detailed local spatial–spectral patterns (feature learning), while CL explicitly optimizes the feature space for separability between distinct instances (instance discrimination). Their synergistic integration in SSupMamba drives substantially more robust and discriminative feature learning than either approach independently.

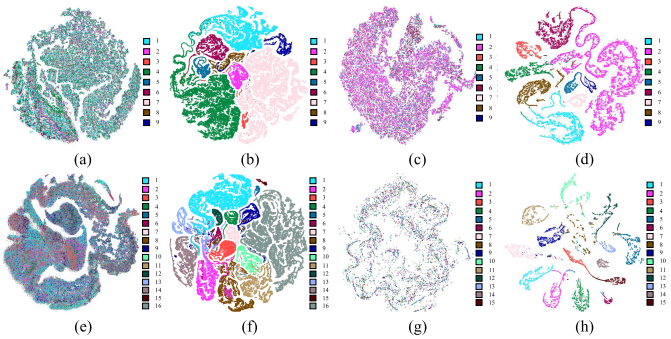


Fig. 11. T-SNE visualization of raw versus SSupMamba features for (a) and (b) LK, (c) and (d) UP, (e) and (f) HC, and (g) and (h) HU datasets.

4) Fine-Tuning Strategies: An ablation study comparing fine-tuning strategies—using View 1, View 2, or both concatenated—reveals generally consistent performance across most datasets (Table VII). Slightly larger variations are observed on HU and HC, where View 1 outperforms View 2 by 3.39% and nearly 2%, respectively. Notably, using both the views together does not consistently improve accuracy and sometimes underperforms a single view. These results suggest that while multiview pretraining learns generally robust features, certain views may capture more discriminative information for specific datasets. Moreover, simple concatenation may not effectively integrate complementary information, indicating potential for more sophisticated fusion methods in future work.

F. Feature Visualization

Fig. 11 visualizes feature distributions via t-SNE [50] to compare the representation quality of the original spectral features and those learned by SSupMamba across all the four HSI datasets. The first row shows results for LK and UP, and the second for HC and HU. SSupMamba consistently improves feature separability, forming well-clustered groups with clear boundaries and greater interclass distance in the embedding space. This improvement is most notable on UP: where raw features show severe overlap (Fig. 11, row 1, second column), SSupMamba produces distinct, separable clusters (Fig. 11, row 1, fourth column). The resulting well-structured features enable effective classification even with linear models.

G. Execution Time and Model Complexity

We evaluate computational efficiency through model parameters and training time (Table VIII). Model parameters reflect spatial complexity and indicate learning capacity. Most deep HSIC methods involve separate training and testing phases. Supervised methods train quickly due to limited labels, while FSL methods such as DCFSL require extensive source-domain pretraining for fast target adaptation. Self-supervised methods, including C-SS-MTr and SSupMamba, also use pretraining before fine-tuning. FSL methods generally incur

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TABLE VI
OA(%) OF SSUPMAMBA WITH DIFFERENT MODULES

Method	CSMB	SpeMask	SpaMask	CL	LK	UP	HC	HU
SAEM(Spectral)	✓	✓	-	-	81.39±1.03	74.24±1.28	69.40±1.07	59.02±0.86
SAEM(Spatial)	✓	-	✓	-	80.30±0.99	72.96±0.79	70.78±1.15	55.38±0.87
SAEM(View 1)	✓	✓	✓	-	87.17±1.60	79.43±1.42	75.87±1.71	66.75±1.75
SAEM(View 2)	✓	✓	✓	-	86.00±2.10	76.49±1.65	75.76±2.31	61.44±1.96
CL(No SAEM)	✓	-	-	✓	86.92±0.83	75.51±1.35	77.25±1.11	72.31±1.08
SAEM+CL(Mamba)	-	✓	✓	✓	90.41±1.29	88.64±1.05	80.08±1.42	81.72±1.19
SAEM+CL(CSMB)	✓	✓	✓	✓	**94.89±1.49**	**91.16±1.64**	**86.94±1.10**	**86.60±1.41**

TABLE VII
OA(%) OF ABLATION STUDY ON FINE-TUNING STRATEGIES

Views	LK	UP	HC	HU
View 1	94.89±1.49	91.16±1.64	**86.94±1.10**	**86.60±1.41**
View 2	**95.41±1.55**	89.67±1.68	84.98±2.48	83.21±2.07
Concate	95.33±1.18	**91.44±1.73**	84.25±1.53	85.31±1.81

TABLE VIII
COMPUTATIONAL COMPLEXITY ANALYSIS OF VARIOUS METHODS

Mthods	Param(K)	Training times(s)	Testing times(s)
PCA+SVM	–	29.23	1.99
SSTN	24.72	2172.81	16.91
DCFSL	4276.39	2600.78	13.93
MambaHSI	435.03	233.84	1.94
SS-MTr	1349.39	1365.84	93.14
C-SS-MTr	1540.69	1410.11	94.50
TCRL	165.89	527.42	2.41
CMLPL	745.81	559.56	14.37
SSupMamba	739.31	458.97	8.68

higher pretraining costs than supervised approaches. Although self-supervised methods also require substantial pretraining, they are slightly more efficient than FSL. SSupMamba uses fewer parameters than DCFSL (FSL) and SS-MTr/C-SS-MTr (self-supervised) and trains faster than leading cross-domain FSL method DCFSL—all while achieving higher accuracy, demonstrating a practical accuracy–efficiency tradeoff for HSI applications.

V. CONCLUSION

This article proposes SSupMamba, a novel framework advancing small-sample HSIC through three key innovations: first, the CSMB enhances complex feature capture via multidirectional scanning. Second, the SAEM module uses joint spectral–spatial masking to learn discriminative representations through challenging reconstruction tasks. Third, SSupMamba synergistically integrates MIM with CL, generating well-separated embeddings that enable state-of-the-art classification performance under extreme sample scarcity (five samples/class). Extensive experiments across four datasets confirm significant accuracy improvements over existing methods while maintaining parametric efficiency. Current limitations include dependence on two-stage training and moderate parameter volume. Future work will develop end-to-end architectures, explore open-set classification paradigms, and investigate multimodal data fusion to further advance resource-efficient hyperspectral analysis.

REFERENCES

[1] M. F. Guerri, C. Distante, P. Spagnolo, F. Bougourzi, and A. Taleb-Ahmed, "Deep learning techniques for hyperspectral image analysis in agriculture: A review," *ISPRS Open J. Photogramm. Remote Sens.*, vol. 12, Apr. 2024, Art. no. 100062.

[2] M. Wang et al., "Tensor decompositions for hyperspectral data processing in remote sensing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 1, pp. 26–72, Mar. 2023.

[3] L. Chen, J. Wu, Y. Xie, E. Chen, and X. Zhang, "Discriminative feature constraints via supervised contrastive learning for few-shot forest tree species classification using airborne hyperspectral images," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113710.

[4] N. Liu, P. A. Townsend, M. R. Naber, P. C. Bethke, W. B. Hills, and Y. Wang, "Hyperspectral imagery to monitor crop nutrient status within and across growing seasons," *Remote Sens. Environ.*, vol. 255, Mar. 2021, Art. no. 112303.

[5] X. Fu, S. Jia, L. Zhuang, M. Xu, J. Zhou, and Q. Li, "Hyperspectral anomaly detection via deep plug-and-play denoising CNN regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9553–9568, Nov. 2021.

[6] A. Nowakowski et al., "Transfer learning in Earth observation data analysis: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 1, pp. 2–33, 2024.

[7] W. Li, S. Prasad, J. E. Fowler, and L. M. Bruce, "Locality-preserving dimensionality reduction and classification for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1185–1198, Apr. 2012.

[8] J. Xia, P. Ghamisi, N. Yokoya, and A. Iwasaki, "Random forest ensembles and extended multiextinction profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 202–216, Jan. 2018.

[9] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[10] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[11] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[12] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[13] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[14] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[15] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[16] M. Zhu, H. Wang, Y. Meng, Z. Shan, and Z. Ma, "Lightweight multi-view mask contrastive network for small-sample hyperspectral image classification," in *Proc. Chin. Conf. Pattern Recognit. Comput. Vis. (PRCV)*, 2023, pp. 478–490.

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[17] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508416.

[18] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[19] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Modeling*, 2023, pp. 1–16.

[20] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, 2024, pp. 62429–62442.

[21] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[22] T. Dao and A. Gu, "Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality," in *Proc. 41st Int. Conf. Mach. Learn.*, 2024, pp. 10041–10071.

[23] T. Huang, X. Pei, S. You, F. Wang, Q. Chen, and C. Xu, "LocalMamba: Visual state space model with windowed selective scan," in *Proc. Eur. Conf. Comput. Vis.*, 2025, pp. 12–22.

[24] A. Shaker, S. T. Wasim, S. Khan, J. Gall, and F. S. Khan, "GroupMamba: Efficient group-based visual state space model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 14912–14922.

[25] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[26] C. Wang, J. Huang, M. Lv, H. Du, Y. Wu, and R. Qin, "A local enhanced mamba network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, pp. 1–13, Sep. 2024, Art. no. 104092.

[27] D. Ibañez, R. Fernandez-Beltran, F. Pla, and N. Yokoya, "Masked auto-encoding spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542614.

[28] L. Scheibenreif, M. Mommert, and D. Borth, "Masked vision transformers for hyperspectral image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2166–2176.

[29] P. Guan and E. Y. Lam, "Cross-domain contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528913.

[30] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[31] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 572–585.

[32] H. Zhang et al., "A survey on visual mamba," *Appl. Sci.*, vol. 14, no. 13, p. 5683, Jun. 2024.

[33] G. Wang, X. Zhang, Z. Peng, T. Zhang, and L. Jiao, "S^2Mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511413.

[34] J. Sheng, J. Zhou, J. Wang, P. Ye, and J. Fan, "DualMamba: A lightweight spectral–spatial mamba-convolution network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501415.

[35] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.

[36] Z. Huang et al., "Contrastive masked autoencoders are stronger vision learners," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 4, pp. 2506–2517, Apr. 2024.

[37] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.

[38] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718.

[39] J. Lin, F. Gao, X. Shi, J. Dong, and Q. Du, "SS-MAE: Spatial–spectral masked autoencoder for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531614.

[40] Y. Tian, D. Krishnan, and P. Isola, "Contrastive multiview coding," in *Proc. Eur. Conf. Comput. Vis.*Cham, Switzerland: Springer, 2020, pp. 776–794.

[41] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

[42] R. E. Kalman, "A new approach to linear filtering and prediction problems," Tech. Rep., 1960.

[43] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[44] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715.

[45] L. Tu, J. Li, X. Huang, J. Gong, X. Xie, and L. Wang, "S2HM2: A spectral–spatial hierarchical masked modeling framework for self-supervised feature learning and classification of large-scale hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517019.

[46] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014.

[47] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[48] X. Zhang, S. Yang, Z. Feng, L. Song, Y. Wei, and L. Jiao, "Triple contrastive representation learning for hyperspectral image classification with noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 500116.

[49] L. Liu, H. Zhang, and Y. Wang, "Contrastive mutual learning with pseudo-label smoothing for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.

[50] L. v. d. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.

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